Highlights

Domain-Validity-Gated Metamorphic Testing of Scientific ML Surrogates

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- A domain-validity rubric screens candidate MRs for SciML surrogate testing.
- MR tolerances are gated by the measuring operator's floor ($O(h)$ for divergence).
- MR cards convert retained candidates into auditable executable test assets.
- A fail-closed claim ledger ties every reported number to a tracked artifact.
- K=6 rosters show where the admitted MR set bounds detector coverage.

Domain-Validity-Gated Metamorphic Testing of Scientific ML Surrogates

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Abstract

Context: Scientific machine-learning (SciML) surrogates approximate expensive simulations, but exact outputs for arbitrary inputs are unavailable (the oracle problem). Metamorphic testing addresses this through relations across executions, yet a candidate relation is not automatically valid: its preconditions, output mapping, and the scoring operator's numerical floor determine whether a violation is meaningful.

Objective: We investigate how physics-derived metamorphic relations (MRs) can be transformed into auditable, executable, and interpretable oracle-free V&V assets for SciML surrogates, making the candidate-to-asset step explicit about the physical, numerical, and software conditions each relation requires, and whether the validity gating changes V&V decisions.

Method: We propose (i) a domain-validity rubric admitting a candidate only when its tolerance dominates the operator's numerical floor and its preconditions hold; (ii) an MR-card executable-asset format recording transformations, mappings, metrics, tolerances, exclusion rules, and relation-level verdicts; and (iii) a case-study protocol on MeshGraphNets cylinder-flow surrogates, with a claim ledger binding every result to an artifact.

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Results: The gate is shown to change V&V outcomes, not merely annotate them: it removes detectors that would raise false alarms on a correct surrogate, and on a second CFD task it rejects, on physical grounds, a continuity relation it only defers on cylinder flow. On cylinder flow the typed verdicts recur unchanged across a same-family roster and a PhysicsNeMo replication (node permutation exact, mirror-y bounded out-of-distribution stress, absolute conservation deferred against a closed-form floor); on PINN and FNO surrogates the gate gives each task's relation-appropriate verdict. Seeded-fault tests bound detector coverage, not as a general fault-detection rate.

Conclusion: Validity-gated MR assets make oracle-free SciML V&V more auditable by recording why a relation is admitted, rejected, deferred, or interpreted as out-of-relation-domain. The study supports a reusable workflow for converting physics-derived relations into executable assets with typed verdicts. Broader generalization is future work.

Keywords: Metamorphic testing, metamorphic relation identification, oracle problem, scientific machine learning, neural surrogates, MeshGraphNets, domain-validity rubric

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1. Introduction

Scientific machine-learning (SciML) surrogates are now used as software components for approximating expensive simulations. Mesh-based neural simulators such as MeshGraphNets predict physical fields on irregular meshes [1], but they also inherit a classical testing difficulty: for many candidate inputs there is no cheap exact output against which the program can be checked, the test oracle problem [2]. Held-out rollout error is therefore necessary, but it does not by itself tell a tester whether the software respects relations that should hold under physically meaningful transformations.

Metamorphic testing (MT) addresses the oracle problem by checking relations among multiple executions rather than a single expected output [3, 4]. Conservation laws, symmetry relations, nondimensional similarity, and continuity conditions can all suggest candidate metamorphic relations (MRs) for scientific software [5, 6, 7]. The testing problem considered in this paper begins after such candidates have been proposed. A physics-derived relation is not automatically a valid software test: mirror symmetry may depend on geometry and boundary labels; a divergence relation may depend on a discrete operator, mesh weights, and boundary treatment; a transformed case may leave the relation's domain before it reveals any SUT inconsistency.

For SciML surrogate software, this creates an interpretability problem for oracle-free V&V. When an MR check fails, the tester must ask whether the SUT violated a genuine relation, the relation was applied outside its validity conditions, the output mapping was incompatible with the software representation, or the scoring operator could not resolve the claimed effect. Treating all such outcomes as ordinary pass/fail MT verdicts makes the evidence hard to audit and easy to overclaim.

This paper therefore reframes SciML MR use as *validity-gated V&V asset construction*. It does not claim to invent MT, and it does not replace residual, uncertainty, accuracy, or trust-region diagnostics. Its contribution is to turn physics-derived candidate relations into auditable assets by combining measurement-floor admissibility, MR-card executable assets, and typed verdict ledgers. The gate is more than organizational: it is shown to change V&V outcomes, removing detectors that would raise false alarms on a correct surrogate and rejecting, on physical grounds, a relation it only defers under different physics. Physical knowledge, expert reasoning, LLM-assisted candidate lists, and the NOETHER two-layer MetaPattern model [8] may suggest candidate relations, but the evidence supports bounded SciML V&V utility rather than general SciML reliability or real-world defect-detection rates.

Two principles organize the method. First, an MR is *admissible* only when its validity conditions hold for the SUT and the verdict is numerically decidable: the tolerance must dominate the intrinsic floor of the measuring operator (machine precision, a mapping floor, or a discretization floor, by relation type). Second, relation-level verdicts are interpreted along two axes, relation-violation and domain-violation magnitude, separating SUT inconsistency from out-of-relation-domain application, numerical-floor deferral, and inconclusive evidence. We refer to the resulting framework as domain-admissibility-gated, relation-indexed SciML V&V. The main empirical setting is MeshGraphNets-family cylinder flow, with the same admissibility predicate additionally exercised on a second CFD task and a second PDE family; these are bounded case studies, not cross-domain reliability estimates.

1.1. Research questions

The main research question is:

RQ0. How can physics-derived metamorphic relations for scientific-machine-learning surrogate software be transformed into auditable, executable, and interpretable oracle-free V&V assets, while distinguishing genuine SUT inconsistency from relation invalidity and numerical measurement limits?

We decompose this into four questions:

RQ1 – Admissibility. What validity conditions determine whether a physics-derived candidate MR is admissible as an oracle-free test for a specific SciML surrogate SUT?

RQ2 – Asset construction. How can admissible MRs be represented as reusable MR cards and executable test assets with explicit transformations, mappings, metrics, tolerances, exclusions, and evidence records?

RQ3 – Verdict interpretation. How can relation-level verdicts separate SUT inconsistency from out-of-relation-domain application, numerical tolerance limits, and inconclusive evidence?

RQ4 – Empirical utility. In bounded SciML surrogate case studies, what evidence does the validity-gated workflow provide beyond rollout accuracy, generic MR generation, and LLM-assisted candidate elicitation?

1.2. Contributions

The single methodological idea is to gate a candidate relation's admissibility on numerical decidability: a relation is admitted only when its verdict tolerance dominates the measuring operator's own intrinsic error floor, a test that to our knowledge is new for learned SciML surrogates evaluated out of distribution. The four scoped contributions below build this idea into an auditable workflow, each tied to a research question and to evidence reported later (Table 3). The positive claim is methodological: a measurement-floor admissibility gate, a typed domain-inadmissibility verdict, and a seeded-fault diagnostic stress test whose MR-class-to-fault-class diagnostic is reported as stress-test evidence rather than a validated localization model.

Measurement-floor admissibility. The method admits a relation only when its verdict tolerance dominates the intrinsic numerical floor of the measurement operator. For the deployed cylinder-flow mesh, the divergence floor is closed form, a leading-order predictor matches the measured floor to within 0.5%, a rigorous a-priori bound dominates it, and the floor is empirically topology-stable across a structured and an unstructured Delaunay mesh (Sec. 5.6); a general unstructured-mesh bound is left to future work.

Validity-gated V&V asset construction. This workflow contribution transforms physics-derived candidate MRs into auditable executable testing assets. Each admitted MR is represented as an MR card and executable runner that records the transformation, mapping, metric, tolerance, exclusion rule, and ledger fields needed for independent inspection.

Typed relation-level verdicts. The verdict scheme separates relation violation from domain violation, instantiating for physics-governed SciML the constraint-architecture pattern behind bug-versus-inapplicability reasoning [9, 10] and MetaTrimmer [11]. The added element is a typed classification of physical, geometric, boundary-condition, mapping, and operator-admissibility limits.

Bounded empirical utility. The case studies evaluate what this workflow adds beyond rollout accuracy, generic MR generation, and LLM-assisted MR candidate elicitation, drawing primarily on MeshGraphNets cylinder-flow runs with supporting airfoil, PINN, and FNO checks. This utility is a demonstrated change in V&V outcomes rather than bookkeeping (Section 5). The admissible MR set is also used to interpret fault-detection boundaries and blind spots, but this coverage reading is a discussion implication of the admitted relations rather than a separate formal research question.

2. Related Work

2.1. Mesh-based neural simulation

Mesh-based neural simulators learn dynamics on graph or mesh representations of physical systems. MeshGraphNets encodes mesh nodes and edges, propagates information by message passing, and predicts future physical fields through autoregressive rollout [1]. In this paper these models are treated as software systems under test (SUTs). The relevant testing question is therefore not only whether a rollout is accurate on held-out trajectories, but whether the SUT preserves declared relations under controlled transformations and explicit tolerances.

2.2. Metamorphic testing and the oracle problem

Metamorphic testing (MT) addresses programs lacking a per-input oracle [3, 4, 12]. Rather than checking each output against an expected value, MT checks a relation among source and follow-up executions. This framing has been applied to scientific software, PDE programs, simulations, and ML systems [13, 14, 5, 7], including recent applications to tabular ML such as credit scoring [15]. For SciML surrogates, however, an MR is not merely an input perturbation: its validity depends on governing equations, boundary conditions, mesh representation, output mapping, and numerical tolerance. This makes MR admissibility part of the oracle construction problem itself.

2.3. MR identification for scientific and simulation software

Testing scientific software without a reliable oracle is a long-standing problem, and metamorphic testing is the most developed response in the surveyed literature [16]. MRs can be elicited from monotonicity, conservation, scaling, and domain-specific expectations [5, 6, 7, 17]. Predictive MR identification has been pursued with graph-kernel learning over scientific source code [18], by separating input and output patterns and by image-based output-pattern recognition for numerical solver codes [19, 20], and with large language models that discover metamorphic-relation variables for scientific software [21]. Symmetry-based MRs paired with statistical violation criteria have also been applied to deep trajectory predictors [22, 23]. Data-driven search can discover candidate transformations in complex physical software [24], and design assumptions can be encoded as MRs for CPS testing under control–plant invariants [25].

Separate work has made MR specification and applicability more explicit. General MT tooling has explored machine-readable MR representations, constraint definition, test-data-driven MR selection, and general MR specification languages [10, 26, 27]. These systems strengthen MR engineering in the large, but they do not by themselves decide whether a physics-derived relation is numerically measurable, in-domain for a transformed CFD case, or interpretable as a SciML V&V verdict.

These approaches are valuable candidate sources, but candidate identification is not the same as admissible SciML test construction. MR identification, including our own prior physics-derived MR construction for numerical solver and nuclear-power software [28, 19], is itself established work; this paper takes candidate MRs as input and contributes the admissibility gate rather than a new identification method. Fault-oriented MR work ranks relations by observed detection power [29, 30]; the present paper uses the admitted relation set later to interpret bounded coverage and blind spots, but treats that as a consequence of admissibility rather than as the primary novelty.

2.4. Physics-based MT for learned scientific simulators

Three contemporary strands are closest to this paper. Yang and Chui [31] and Reichert et al. [32] applied physics-derived MRs to hydrologic ML or conceptual models, showing that learned-model accuracy and relation consistency can diverge and that calibration or forcing regimes affect interpretation. Eniser et al. [33] introduced *relaxations*, numerical tolerances embedded in MR oracles, for stochastic RL policy testing. Duque-Torres et al. [9, 10, 26] and MetaTrimmer [11] address bug-versus-inapplicability reasoning with explicit MR constraints and automated constraint derivation.

The present work combines these ideas in a SciML-specific V&V workflow. From physics-based MR testing it takes relation-level checks; from relaxations it takes the need for calibrated tolerances; from violation-attribution work it takes the need to avoid treating inapplicability as a SUT failure. What none of these strands supplies is the element doing the load-bearing work here: admissibility is made contingent on numerical decidability. Relaxations attach a tolerance to an oracle, and constraint- or attribution-based work decides whether a relation applies to an input, but neither requires the verdict tolerance to dominate the measuring operator's intrinsic error floor, a property of the discretization fixed before any fault or SUT is considered. That floor gate, not the recording discipline, is what is genuinely new here. The admissibility decision, tolerance rationale, executable asset, and typed verdict are then recorded together and tied to evidence artifacts. Table 1 positions the paper against the closest-prior capabilities.

Table 1: Closest-prior capability matrix. The table positions the paper against capabilities needed by RQ0–RQ4; it does not assert that prior work should have solved the SciML-specific combination.

Approach	What it supplies	What remains missing for RQ0–RQ4	What this paper adds
Scientific-software MR identification	Candidate relations from domain rules, source-code learning, LLM variables, and data-driven search	Candidate validity, numerical decidability, executable asset format, and typed SciML verdicts	Treats these methods as candidate sources, not validity authorities
Reichert et al.	Physics-derived MRs and informal filtering in a learned hydrologic surrogate	Explicit admissibility predicate, operator-floor tolerance, reusable MR cards, and typed verdicts	Formalizes the candidate-to-verdict path for mesh-based SciML surrogates
Eniser et al.	Relaxed MR oracles with numerical thresholds	Measurement-operator floor and relation-domain interpretation for deterministic SciML surrogates	Grounds tolerance in the scoring operator's own numerical floor
Duque-Torres / MetaTrimmer	Binary bug-versus-inapplicability separation with explicit MR constraints and automated constraint derivation	A typed inadmissibility verdict and operator-floor admissibility for physics-governed SciML	Replaces the binary skip/proceed pre-filter with a two-axis typed verdict tied to physical, geometric, boundary, and operator limits
Formal MR specification and constraint tooling	Machine-readable MR descriptions, constraint phases, and supporting systems	Physics-specific admissibility dimensions, measurement floors, and executable evidence ledgers for SciML	Uses specification ideas but binds each verdict to relation-domain and numerical-decidability evidence
This paper	Physics, expert, and LLM-assisted candidates plus artifact-backed execution	Bounded case-study evidence, not general SciML reliability or baseline superiority	Constructs MR cards, runners, ledgers, and typed verdicts under a four-condition gate

2.5. SciML V&V, residuals, UQ, and failure modes

SciML reliability research has developed complementary tools: physics residuals, uncertainty quantification, conformal prediction, certified error bounds, and failure-mode analysis [34, 35, 36, 37, 38]. PINN training can exhibit gradient-flow pathologies that bias trained models away from the governing residual [39, 37], sharpening the need for checks that do not assume a converged surrogate. These diagnostics are not competitors to MT. A residual, equivariance error, or uncertainty score becomes an executable MR only when paired with a source case, follow-up transformation, tolerance rule, exclusion rule, and verdict interpretation. The proposed workflow uses such diagnostics as possible measurements inside a multi-execution oracle-free asset.

2.6. Hybrid ML-solver trust regions

Hybrid ML-solver frameworks use residual thresholds or error estimates to decide when a learned component should be trusted [40, 41]. Such systems operationalize runtime trust regions, whereas this study acts offline: physically derived transformations are applied before deployment to estimate where relation violations occur.

The distinction from residual- and uncertainty-based trust estimation is deliberate. UQ, conformal prediction, residual-threshold trust regions, and physical-consistency diagnostics that score a physically derived residual on observed outputs [42] locate unreliable behavior from observed inputs. The present method acts in relation space: it applies a controlled transformation and reports which necessary relation breaks under it, indexed by that transformation. The claim is complementarity, not superiority: relation-indexed evidence answers a different question from scalar accuracy, residual magnitude, or uncertainty.

2.7. What is new and what is not new

Metamorphic testing, MR identification, scientific-software MT, residual diagnostics, uncertainty quantification, LLM candidate generation, and the NOETHER two-layer MetaPattern model are established or emerging sources of testing ideas. The contribution here is narrower: SciML MR use should be treated as a domain-validity and asset-construction problem. A candidate relation becomes a useful oracle-free V&V asset only after its physical basis, transformation preconditions, output mapping, tolerance, exclusion rule, executable artifact, and relation-level verdict are recorded. Recent work derives or refines domain-specific MRs for scientific programs directly from their numerical models [43, 44]; the wedge here is the learned surrogate evaluated out of distribution and the requirement that a relation's tolerance dominate the measuring operator's numerical floor, rather than testing a numerical solver against derived relations. The missing combination is thus a validity-gated, executable, and auditable workflow for physics-governed SciML surrogate software (Table 1).

3. Method

3.1. Overview

The method answers RQ1–RQ3 by turning a physics-derived candidate relation into an auditable V&V asset only after three checks have been made explicit: admissibility, asset construction, and verdict interpretation. Figure 1 summarizes the workflow. Candidate generation is deliberately separated from validity judgment: expert reasoning, physical laws, the NOETHER two-layer MetaPattern model, and LLM-assisted lists may propose relations, but none certifies that a relation is valid for a specific SUT.

Terminology and verdict types. A verdict is read on two axes, relation-violation and domain-violation magnitude (detailed in Sec. 3.5). A candidate relation is accordingly *admitted* (a large in-domain violation then reads as SUT inconsistency), *rejected* on physical grounds, *deferred* when no tolerance dominates the operator floor, marked *out-of-relation-domain*, or left *inconclusive*. Four MR families recur in the case studies: node-permutation equivariance, mirror-y reflection symmetry, discrete-divergence continuity/conservation, and exact mirror-y on a constructed symmetric mesh.

3.2. Candidate relation sources: a two-layer MetaPattern model

We organize candidate relations with the two-layer MetaPattern model of NOETHER [8], a same-author preprint, which grounds each relation in an explicit algebraic generating structure, superseding the earlier descriptive three-level physical/computational/code classification [45]. This two-layer organization is an optional scaffold for generating and naming candidate relations; the load-bearing element of the method is the admissibility gate of Section 3.3, which applies however a candidate was sourced.

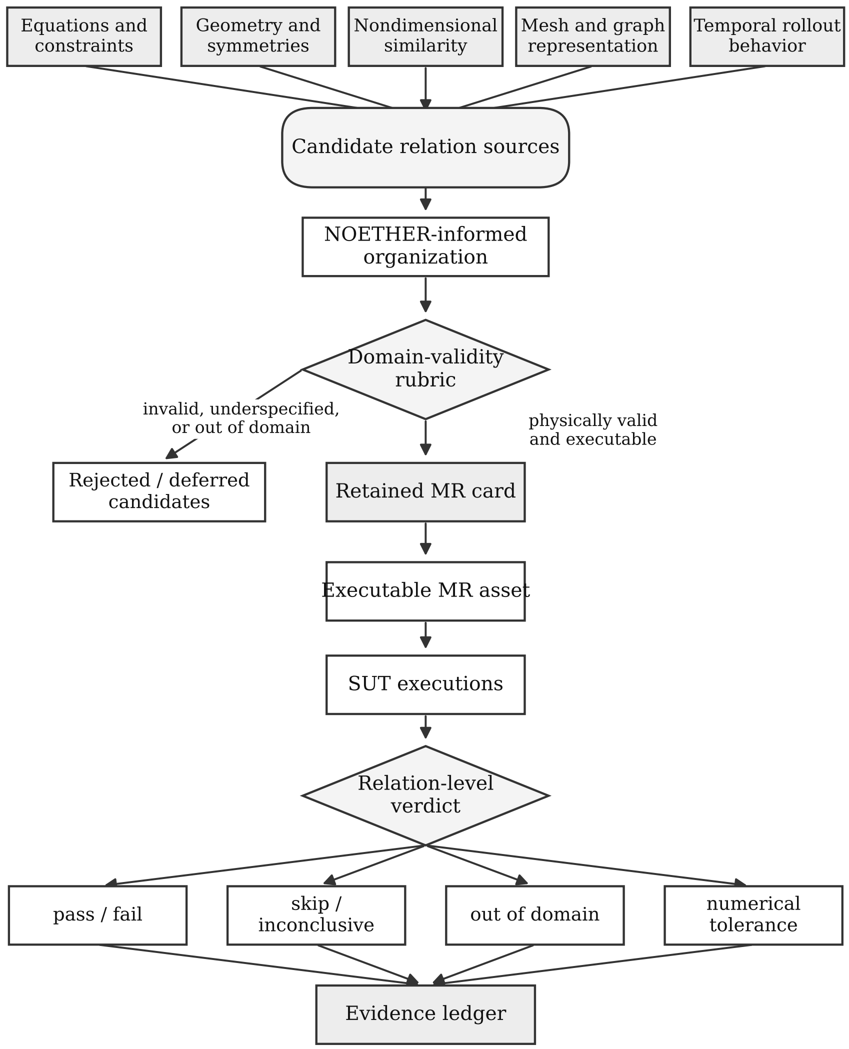

Figure 1: Validity-gated V&V workflow in which gate-admitted candidate relations become MR cards, executable runners, typed relation-level verdicts, and evidence ledgers.

Layer 1 (MetaPattern). A *MetaPattern* is the equivalence class of metamorphic relations generated, through TRANSLATE, from one minimal algebraic structure of the program-induced operator algebra $\mathcal{A}_P$ (its generating basis). The five MetaPatterns $m_{\text{inv}}, m_{\text{mono}}, m_{\text{adj}}, m_{\text{rev}}, m_{\text{conv}}$ are generated respectively by a group action (G), a partial order ($O_{\leq}$), a self-adjoint operator (T^*), a time-reversal involution ($\mathcal{T}^*_{\text{rev}}$), and a parametrized limit ($\mathcal{L}^*$); their set is $\mathbb{M}(\mathcal{A}_P)$. By the Noether-style correspondence between symmetries and conserved quantities, a conservation relation is a member of the group-action MetaPattern m_{inv} rather than a separate structure.

Layer 2 (MR family). An *MR family* is the set of executable relations obtained by instantiating one MetaPattern under a fixed transformation template and a chosen mode (input-orbit or implementation-orbit). Families are written in the sans-serif form $\mathsf{f}_{\text{parent.child}}$, where the stem before the dot names the parent MetaPattern; the legacy enumeration letters a–j are kept in parentheses as cross-reference anchors. The map from MetaPatterns to families is one-to-many: $G \to \{\mathsf{f}_{\text{inv.eqv}}, \mathsf{f}_{\text{inv.con}}\}$ (equivariance a, conservation b), $T^* \to \{\mathsf{f}_{\text{adj.self}}, \mathsf{f}_{\text{adj.dual}}\}$ (self-adjoint c, adjoint-duality d), $\mathcal{T}^*_{\text{rev}} \to \{\mathsf{f}_{\text{rev.traj}}\}$ (trajectory-reversal e), $O_{\leq} \to \{\mathsf{f}_{\text{mono.stat}}, \mathsf{f}_{\text{mono.shape}}\}$ (static-order f, dynamic-shape g), and $\mathcal{L}^* \to \{\mathsf{f}_{\text{conv.lim}}, \mathsf{f}_{\text{conv.rate}}, \mathsf{f}_{\text{conv.repr}}\}$ (convergence h, accuracy-order i, representation-invariance j).

For the cylinder-flow surrogates studied here, the instantiated families are $\mathsf{f}_{\text{inv.eqv}}$ (equivariance, a), mirror-y reflection; $\mathsf{f}_{\text{inv.con}}$ (conservation, b), discrete-divergence continuity and, on the airfoil task, compressible mass balance; $\mathsf{f}_{\text{conv.repr}}$ (representation-invariance, j), node-permutation and encoding contracts; and $\mathsf{f}_{\text{conv.lim}}$ and $\mathsf{f}_{\text{conv.rate}}$ (convergence, accuracy-order, h–i), the operator-floor mesh-refinement relation. The $\mathsf{f}_{\text{mono.stat}}$ (static-order, f) Reynolds–Strouhal monotonicity is a candidate not exercised here, $\mathsf{f}_{\text{rev.traj}}$ (trajectory-reversal, e) is empty by construction for viscous dissipative flow, and the self-adjoint families $\mathsf{f}_{\text{adj.self}}$ and $\mathsf{f}_{\text{adj.dual}}$ (c–d) appear only in the cross-family PINN and FNO extensions. The same relation may be admissible as one family yet inadmissible once instantiated for a given SUT: mirror symmetry holds on a symmetric domain but not on an asymmetric mesh with incompatible boundary labels, and conservation is a physical expectation but not an executable verdict when the discrete measuring operator has an unresolved floor. The admissibility gate records where each relation survives.

3.3. Admissibility gate

A candidate relation is an *admissible MR* only when four conditions hold together. First, it must have a physical or software basis. Second, its transformation preconditions must hold for the source and follow-up cases. Third, boundary conditions and output mappings must remain compatible after transformation. Fourth, the verdict must be numerically decidable: the tolerance must dominate the intrinsic error floor of the measuring operator. None of these conditions is optional. Table 2 records the gate, the MR-card fields, and the verdict semantics.

Table 2: Admissibility gate, MR-card fields, and verdict semantics. This is the method table for RQ1–RQ3.

Method element	**Required record**	**Allowed interpretation**	**RQ**
Basis	Equation, boundary condition, nondimensional law, representation property, numerical assumption, or implementation contract	Relation may be considered only for SUTs where this basis applies	RQ1
Preconditions	Source-case constraints, follow-up transformation, preserved quantities, and exclusion rule	Missing preconditions produce skip or out-of-relation-domain, not SUT failure	RQ1
Mapping compatibility	Boundary-label compatibility and output mapping from follow-up output back to source coordinates/fields	Incompatible mapping rejects or downgrades the candidate	RQ1
Numerical decidability	Metric, tolerance, tolerance provenance, and measurement-floor evidence	A verdict is meaningful only when tolerance dominates the operator floor	RQ1
MR card	Identifier, source category, transformation, mapping, metric, tolerance, exclusions, expected verdicts, and artifact schema	The admitted relation becomes a reusable executable V&V asset	RQ2
Evidence ledger	Source/follow-up run ids, metric outputs, floor evidence, exclusions, logs, and verdict	The reported claim is auditable from artifacts rather than prose alone	RQ2
Verdict semantics	Pass, fail, skip, out-of-relation-domain, numerical-tolerance issue, or inconclusive	Only fail inside the valid domain may be read as SUT inconsistency	RQ3

The gate produces four design-time decisions: admit as an executable

MR; downgrade to OOD-stress when the relation is physically informative but outside its exact domain; reject when basis, preconditions, or mapping fail; or defer when missing numerical evidence prevents a meaningful verdict. For example, node permutation can be admitted as a representation MR when the adapter and inverse mapping are deterministic; mirror-y is conditional on compatible geometry, boundary labels, and vector-component mapping; absolute discrete divergence is deferred when the reference field and measuring operator floor cannot support an absolute verdict.

3.4. MR card and executable asset format

A retained MR is represented as an MR card. The card records the fields listed in Table 2: identifier, source category, physical or software basis, source-case schema, follow-up transformation, preconditions, boundary-condition compatibility, output mapping, metric, tolerance and provenance, expected verdict classes, exclusion rules, artifact schema, and interpretation limits. The card is the reusable design artifact; the executable runner is the operational artifact generated from it.

Each executable asset includes transformation code, runner configuration, metric computation, verdict logic, and artifact recording. The ledger is part of the method rather than only replication packaging: every relation-level claim must be traceable to the source run, follow-up run, mapping, metric output, floor evidence, exclusion decision, and verdict.

3.5. Relation-level verdicts

An MR execution can produce: **pass** (relation holds within tolerance); **fail** (violated within the valid relation domain); **skip** (required precondition missing); **out-of-relation-domain** (the transformed case is outside the relation domain); **numerical-tolerance issue** (the measured effect is dominated by threshold or resolution uncertainty); or **inconclusive** (insufficient artifact).

These verdicts form the two-dimensional space in Figure 2. One axis is relation-violation magnitude, typically expressed as a violation-to-floor or violation-to-tolerance ratio. The other is domain-violation magnitude, meaning how far the transformed case lies outside the relation's validity domain. Low domain violation with high relation violation is the region that may be read as SUT inconsistency; high domain violation is out-of-relation-domain; a violation inside the error floor is a numerical-tolerance issue. This decomposition is the method's answer to RQ3.

In this study, the domain-violation coordinate is operationalized per relation from committed measurements. For mirror-y, $D = m/(m+1)$, where m is the worst reflected-node placement error in median-edge-length units. The synthetic symmetric mesh scores $D \approx 0$, whereas the real asymmetric evaluation mesh scores $D = 0.51$. Other MR families use their own precondition measurements. These D values denote structural placement within each relation's own validity domain: D is a per-relation normalized coordinate, not a cross-relation calibrated metric, and the values cannot be averaged or ranked across MR families; they are read per relation, not as a calibrated boundary measurement. A cross-relation calibration would need to normalize each family by its own admissibility boundary, mapping every gate threshold to a common $D_{\mathrm{cal}} = 1$; that calibration is future work.

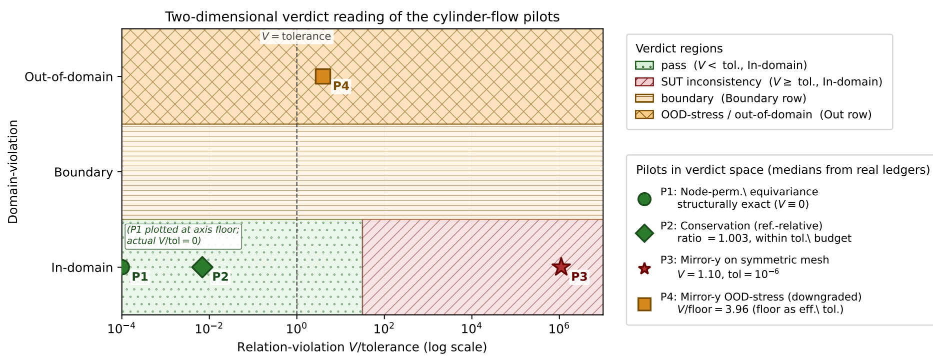

Figure 2: Two-dimensional verdict space plotting relation-violation magnitude against domain-violation magnitude, where only a large in-domain violation reads as SUT inconsistency and the horizontal axis is a per-relation, not cross-family-calibrated, placement.

Retained MRs are interpreted through their two-layer position (Sec. 3.2): a relation's MetaPattern and MR family fix which algebraic invariant it measures, and therefore which faults it can and cannot surface. This is an interpretation protocol, not a validated localization model; the seeded-fault results later in the paper are a bounded stress test of it.

4. Experimental Design

4.1. Evaluation logic

The experiments evaluate whether the method produces auditable, executable, and interpretable MR evidence under bounded conditions. The relations under evaluation are the cylinder-flow MR families of Section 3.2: equivariance ($\mathsf{f}_{\text{inv.eqv}}$) and conservation ($\mathsf{f}_{\text{inv.con}}$) under m_{inv}, representation-invariance ($\mathsf{f}_{\text{conv.repr}}$) and the convergence/accuracy-order pair ($\mathsf{f}_{\text{conv.lim}}$, $\mathsf{f}_{\text{conv.rate}}$) under m_{conv}; the static-order ($\mathsf{f}_{\text{mono.stat}}$) and self-adjoint ($\mathsf{f}_{\text{adj.self}}$, $\mathsf{f}_{\text{adj.dual}}$) families are exercised only as candidate or cross-family checks, and trajectory-reversal ($\mathsf{f}_{\text{rev.traj}}$) is empty by construction. The design separates four evidence tiers, defined next.

4.2. Subject systems and evidence tiers

Primary evidence comes from a MeshGraphNets-family cylinder-flow case study: a trained checkpoint, its K=6 checkpoint roster over three official held-out trajectories, same-domain S4/S5 wider/deeper variants, a PointMLP coordinate network, and the NVIDIA PhysicsNeMo `MeshGraphNet` production implementation evaluated on official DeepMind cylinder_flow TFRecords. The primary unit is a relation-level cell: MR card, SUT/checkpoint, trajectory or source case, transformed follow-up case, metric, and verdict.

Supporting evidence checks whether the same admissibility predicate can be applied outside the primary mesh-surrogate setting: a K=15 PINN roster and trained FNO-2D checkpoints over 2D Burgers and heat data.

Secondary evidence includes LLM-simulated expert MR design, generic MR-generation scope contrasts, LLM-assisted candidate generation, rollout-accuracy diagnostics, and external witness reruns from the read-only Minimum-MR-SubSet repository.

Stress-test evidence uses seeded-fault and cross-program checks to interpret where the admitted MR set can and cannot observe faults.

4.3. Baselines and comparators

Rollout accuracy, residual/UQ diagnostics, expert-designed candidates, generic MR generators, and LLM-assisted candidates are used to contextualize RQ4. Rollout accuracy asks whether a prediction is close to observed held-out data; an MR asks whether a declared relation holds under a controlled transformation. LLM and generic MR outputs are treated as candidate sources; the admissibility gate, not the generator, decides whether a candidate becomes an executable MR.

4.4. RQ-to-evidence map

Table 3 maps the research questions to evaluation evidence, artifacts, units of analysis, and claim boundaries. It is a design table rather than a result table.

Table 3: Evaluation design by research question.

RQ	Evaluation question	Evidence tier and unit	Artifacts	Claim boundary
RQ0	Does the full workflow connect candidate, gate, executable asset, verdict, and evidence?	Primary workflow trace; MR-card cell	MR cards, runners, claim ledger	Asset construction only; no general reliability claim
RQ1	Which candidates are admissible, rejected, downgraded, or deferred?	Primary and supporting gates; candidate/MR	Rubric records, operator-floor sweep, exclusions	Physical validity is relation- and SUT-specific
RQ2	Are admitted relations executable and auditable?	Primary execution cells; source/follow-up pair	Transformations, runner configs, metric files, logs	Execution completeness, not complete automation of MR discovery
RQ3	Do verdicts separate SUT inconsistency from domain and numerical limits?	Verdict ledger; MR-by-case cell	Two-axis verdicts, metric/floor ratios, exclusion logs	Per-relation interpretation; no cross-relation calibrated domain score
RQ4	What evidence is added beyond rollout accuracy and candidate-generation comparators?	Primary, supporting, secondary, and stress-test tiers	Rollout diagnostics, LLM/generic baselines, witness reruns, seeded-fault ledgers	Complementarity and bounded utility; no baseline superiority
Impl.	What blind spots follow from the admitted MR set?	Stress-test tier; mutant/MR cell	Seeded-fault catalogue and cross-SUT removal check	Bounded coverage interpretation, not real-world detection rate

4.5. Efficacy parameters and statistical plan

Primary parameters are admissibility decision, executable MR rate, verdict distribution, violation magnitude relative to tolerance or floor, exclusion rate, and relation-level interpretation yield. Secondary parameters include inter-rater agreement for candidate screening, flakiness rate, complementarity with rollout accuracy, and stress-test detection by seeded-fault class.

The study reports inferential statistics only where the sampling structure supports them and labels descriptive counts otherwise. Binary detector outcomes over the unified fault catalogue carry Wilson 95% intervals per detector; continuous violation magnitudes use exact Wilcoxon signed-rank tests with a rank-based effect size (Cliff's δ), where the design supports paired comparison; roster medians carry paired bootstrap intervals; and the operator-floor sweep reports a log-log slope with a 95% confidence interval. Repeated cells from one trajectory or one checkpoint family are reported as descriptive cell summaries, not independent population samples.

4.6. Ethics, integrity, and reproducibility

The study does not involve human subjects or personal data. All SUT versions, datasets, checkpoints, scripts, and run logs are recorded when licensing permits. Failed, skipped, out-of-relation-domain, numerical-tolerance, and inconclusive outcomes remain in the ledger. LLM use is restricted to candidate generation and evidence organization, not validity judgment. No candidate MR is described as valid without a passing gate decision and MR card; no OOD violation is described as a program fault without supporting verdict evidence.

5. Results Analysis and Discussion

This section reports the evidence by research question rather than by execution chronology, following the RQ-to-evidence mapping of Table 3. The cylinder-flow case study is the primary evidence; the airfoil, PINN, FNO, and cross-program runs are supporting or secondary; and fault-detection and cross-SUT coverage are discussed only as a bounded implication of the admitted MR set. Where a baseline contrast is reported (Appendix Appendix C), it measures the admissibility gate's own value rather than ranking methods: on the correct SUT a gate-admitted detector raises no false alarm while every gate-rejected template does, and against a rollout-accuracy monitor the MR suite is complementary rather than superior. The secondary baselines, the external-scope audit, and the cross-family PINN and FNO executions that test whether the admissibility predicate is family-agnostic are reported in Appendix Appendix C.

5.1. Claim-to-evidence map

The compact claim-to-evidence map binds every claim to a tracked artifact using reviewer-facing claim names; the full table is given in Appendix A (Table A1), and the authoritative runtime mapping remains in `claim-ledger.yml`.

5.2. MR-card-to-verdict map

Table 4 maps each retained MR card to its rubric decision, runtime verdict, and the interpretation boundary that verdict licenses.

Table 4: MR-card-to-verdict map for RQ1–RQ3.

MR card	Rubric decision	Runtime verdict	Interpretation boundary
Node permutation equivariance	Retained as representation MR	pass sanity; relative L2 = 0.0	Supports one executable representation path; does not establish model reliability.
Mirror-y equivariance (asymmetric eval mesh)	Exact relation out-of-relation-domain; downgraded to approximate OOD-stress	S0 pilot: 10/10 fail; primary upgrade: 180/180 fail across K=6 x 3 trajectories x 10	Bounded within-family OOD-stress violation across held-out trajectories; not by itself exact symmetry, cross-SUT, or geometry-independent evidence.
Mirror-y equivariance (synthetic symmetric mesh)	Exact relation admissible (bijection verified, offset $< 10^{-12}$, type-match 1.0)	S0 pilot: fail, rel L2 1.10; primary upgrade: 18/18 fail across K=6 x 3 input seeds	Exact-symmetry failure where the relation is admissible; synthetic no-obstacle OOD meshes, not accuracy or cross-SUT evidence.
Discrete divergence / conservation	Absolute mass-conservation MR deferred; reference-relative diagnostic retained	S0 pilot inconclusive: reference-relative non-regression guard; primary upgrade: 162/162 pass across K=6 x 3 trajectories x 9	Reference-relative diagnostic only; the absolute conservation relation remains deferred.

MR card	Rubric decision	Runtime verdict	Interpretation boundary
MGN S4/S5 variant workflow	Node permutation admitted; mirror OOD/ conservation diagnostic/exact-symmetry decisions recorded	2/2 node-permutation passes; 60/60 mirror OOD failures; 54/54 conservation-diagnostic passes; 6/6 exact-symmetry failures	Same-domain MGN variant evidence, not an external SUT family.
PointMLP cylinder workflow	Node permutation admitted; mirror OOD/ conservation diagnostic/exact-symmetry decisions recorded	9/9 node-permutation passes; 10/10 mirror OOD failures; 9/9 conservation-diagnostic passes; 3/3 exact-symmetry failures	Different non-MGN cylinder SUT; not PhysicsNeMo/EchoWave or production CFD evidence.
PhysicsNeMo MGN Object-A smoke workflow	Node permutation admitted; mirror OOD/ conservation diagnostic decisions recorded	Node permutation passes on the smoke subset (relative L2 0.0); rollout, mirror OOD, and conservation diagnostic ledgers recorded	Production-framework artifact-chain smoke evidence only; no full-scale production pass/fail rate and not external-aerodynamics evidence.

MR card	Rubric decision	Runtime verdict	Interpretation boundary
FNO periodic translation and conservation	Translation admitted; periodic discrete-conservation MR admitted-with-reference-floor; Dirichlet translation rejected	FNO primary workflow upgrade: 24/24 translation passes, 24/24 conservation failures, and 6/6 rejected Dirichlet exact-MR executions	Full rubric-to-verdict FNO evidence with raw source/follow-up outputs and per-case ledgers; outside cylinder-flow and broad neural-operator claims.

5.3. RQ1–RQ4 primary cylinder-flow evidence

All pilots run on one real trained MeshGraphNets cylinder-flow surrogate, using the read-only Minimum-MR-SubSet checkpoint recorded in the experiment ledger. They exercise three different rubric outcomes; raw outputs, manifests, and metric ledgers are committed (Table A1).

Representation MR. Node-permutation equivariance holds to machine precision (relative L2 = 0.0 at tolerance 1e-6). This is a structural property of message-passing and is reported as a pipeline-implementation sanity check rather than model capability or accuracy evidence.

Geometric MR. The rubric classifies exact mirror-y equivariance as out-of-relation-domain for this mesh (the reflection is non-bijective, the worst reflected-node mismatch is about one edge length, and the cylinder is off-centre by 7.2 mm) and downgrades it to an approximate nearest-neighbour OOD-stress probe scored against a same-space mapping-error floor. The probe failed on 10 of 10 recorded eval frames (median relative L2 0.737, median V/floor 3.96, floor range 3.02–5.55x), with no frame skipped or inconclusive. This is a bounded within-SUT frame-level OOD-stress result: one SUT, one checkpoint, one MR, one eval trajectory. The frames are consecutive, not independent samples, so the exact binomial 95% interval for 10/10 is wide ([0.69, 1.00]). The mapping-error floor derives from the same geometric mismatch that triggered the downgrade, so V/floor alone cannot separate a model-level violation from an amplified geometric artifact on this mesh. The symmetric-mesh run below resolves that ambiguity.

Continuity MR. A P1 discrete-divergence operator yields a non-negligible divergence even for the ground-truth field on this coarse mesh, so an absolute divergence-free tolerance is not calibratable and the absolute mass-conservation relation stays deferred. As a reference-relative diagnostic, the surrogate's predicted next-state divergence stays within about 0.4–0.8% of the reference on the recorded eval frames; the interior-only ratio confirms this is not only a boundary-imposition artifact. This asserts no absolute conservation. Two caveats bound the diagnostic. First, the reference divergence is not yet decomposed into discrete-operator error, solver projection artifact, or genuinely non-solenoidal training data, so if the reference field is itself materially non-solenoidal the reference-relative ratio is a non-regression guard rather than a conservation measurement. Second, the diagnostic uses a 50% regression threshold (ratio > 1.5) on two eval frames only, so "passes" means "does not regress conservation by more than 50% on those frames," not "conserves mass."

Rollout-accuracy diagnostic. On the same eval trajectory, the surrogate's one-step next-state prediction error ($v_{\text{pred}} = v_t +$ denormalized predicted delta, the trainer's own convention) has median relative L2 0.0216 (mean 0.044; min 0.0116, max 0.0788) over the nine recorded transitions. The per-transition error is bimodal, so we quote both statistics. The mirror-y OOD-stress violation (median 0.737) is about 34 times the median accuracy and 17x the mean. The quantities are dimensionless relative L2 values of different objects, so this is an order-of-magnitude gap rather than a precise factor; it is one-step, not a free-running rollout-stability result.

Exact mirror-y on a symmetric mesh. We built a provably symmetric synthetic structured channel mesh whose reflection is a verified bijection (node-type match 1.0, reflection offset $< 10^{-12}$, edge set invariant), so the predicate admits exact mirror-y. On one constructed input the surrogate fails this exact equivariance test (relative L2 1.10). The normalizer control changes the violation only from 1.1032 to 1.1014, so the violation is dominated by learned message-passing weights. The boundary is synthetic, no-obstacle OOD geometry: the result confirms missing exact equivariance but is not a calibrated in-distribution magnitude.

Seeded-fault detection. We re-implemented, from the read-only Minimum-MR-SubSet witness taxonomy, an injected-fault catalogue of 10 pipeline faults across five fault classes (boundary-condition, mesh-adjacency, normalization-scale, temporal-rollout, physical-channel), and used the paper's own MRs as detectors on the model's predicted update. The continuity MR detected the two boundary-condition faults and the gross normalization fault (divergence ratio 3.8–10.6 vs the 1.5 threshold); the symmetry MR detected a physical-channel and a mesh-adjacency fault (violation rising 69–142% above baseline); node-permutation equivariance detected none, because these faults preserve node-relabeling invariance. 5 of 10 mutants were detected by at least one MR, and the detections separate by MR family (continuity to boundary/scale faults, symmetry to physical-channel/adjacency faults), a first bounded test of the interpretation protocol of Section 3.5, suggestive rather than a validation. Three boundaries: the detected faults are gross corruptions that any divergence- or symmetry-sensitive detector would catch; the edge-drop fault is a near-miss (32% mirror-y change vs the 50% threshold) and the boundary faults are invisible to mirror-y by construction; and the remaining undetected faults shift the absolute output without crossing the scored-quantity thresholds, delimiting where these MRs are structurally insensitive. Against a rollout-accuracy monitor on the same faults the two are complementary: the edge-permutation and channel-swap faults the symmetry MR catches leave the rollout error within 1.3× of its in-distribution baseline (0.0216, under a 2× regression threshold), so the MRs surface relation violations an accuracy monitor leaves within band, while the gross boundary/normalization faults trip both and no fault is caught by accuracy alone (claim C38). It is one SUT, one checkpoint, one injected-fault catalogue.

The remaining subsections carry the same evidence without adding another main-text table: the MGN roster tests denominator robustness, the operator-floor sweep calibrates numerical decidability, and the fault catalogue bounds detector geometry. Whether the gate transfers beyond the original MeshGraphNets workflow is tested by the cross-family PINN and FNO executions reported in Appendix Appendix C.

5.4. RQ4 same-task replication across checkpoints, trajectories, and architectures

The single-checkpoint evidence is replayed on a K=6 MeshGraphNets roster and three official DeepMind held-out test trajectories acquired through the Minimum-MR-SubSet loader. S0 reuses the pilot checkpoint, so this is a replication/variant check rather than six independent SUTs. Node permutation is exact on all checkpoints. The primary empirical scope upgrade gives a K=6 x 3 trajectories x 10 mirror-y OOD-stress grid with 180/180 failures, a K=6 x 3 trajectories x 9 conservation-transition grid with 162/162 reference-relative passes while absolute conservation remains deferred, and a K=6 x 3 exact-symmetric-mesh input grid with 18/18 failures. The upgrade removes the single-source-trajectory denominator while remaining within one architecture family and dataset, so per-cell counts are descriptive summaries, not independent-trial inference, with the K=6 checkpoint roster rather than the per-cell denominator as the effective independent unit.

Same-task multi-architecture replication. Three further cylinder-flow executions test whether the verdict pattern is an artifact of one implementation. A same-domain S4/S5 wider/deeper MeshGraphNet variant package reproduces the pattern (2/2 node-permutation passes, 60/60 mirror OOD failures, 54/54 conservation-diagnostic passes, 6/6 exact-symmetry failures). A newly trained PointMLP coordinate network (a different architecture class with no message passing) reproduces it as well (9/9 node-permutation passes, 10/10 mirror OOD failures, 9/9 conservation-diagnostic passes, 3/3 exact-symmetry failures, median rollout relative L2 0.0298). Third, the NVIDIA PhysicsNeMo `MeshGraphNet` production implementation, trained and evaluated on official DeepMind cylinder_flow TFRecords, reproduces the node-permutation pass and the mirror-y OOD-stress failure within a production framework rather than a research codebase. The same admissibility predicate, applied unchanged, produced the same typed decisions on all three; this replicates the verdict pattern across architectures on one task and dataset, and asserts no cross-dataset rate.

5.5. *RQ1/RQ3 second-task gate discrimination: compressible airfoil flow*

The strongest test of a domain-validity gate is whether it produces the correct *different* verdict when the physics changes. We apply the identical four-condition predicate to a genuinely second CFD task on a second official dataset: the DeepMind **airfoil** benchmark (SU2-simulated compressible transonic flow, 5,233-node meshes with density, pressure, and velocity), distinct from incompressible cylinder flow in both physics and data. The same official PhysicsNeMo `MeshGraphNet` (2.33M-parameter official configuration: hidden width 128, 15 processor steps) is trained for 40 epochs on 100 official airfoil train trajectories (GPU-trained to a converged final loss near 0.14) across a K=6 checkpoint roster and evaluated per-trajectory on 40 official test trajectories, yielding 240 (checkpoint, trajectory) pairs. The predicate produces a *different typed inadmissibility structure*, and the difference is the result. **Node-permutation equivariance is admitted and exact** (240/240, relative L2 0.0, training-independent representation contract). **Incompressible divergence-free continuity is rejected at the physical-basis gate** (condition i) because the density varies by a median factor of 2.13x across the field, so $\nabla \cdot u = 0$ is physically false here, the same relation that on cylinder flow passes conditions (i)–(iii) and is only *deferred* at condition (iv), so the gate assigns a categorically different verdict *type* for the physically correct reason on each task. **Compressible unsteady mass conservation** $\partial_t \rho + \nabla \cdot (\rho u) = 0$ is the physically-correct replacement, its absolute discrete form deferred (uncalibratable operator floor) and a reference-relative diagnostic recorded (median residual ratio 1.01 across the K=6 roster). **Mirror-y chord symmetry is rejected at the boundary-precondition gate** (condition iii) because the SU2 trajectories carry a non-zero angle of attack. The one-step rollout error remains high at this training budget (median relative L2 0.92, near the signal magnitude) and is reported only as a training-state diagnostic; notably, none of the four typed verdicts depends on rollout quality: the admission rests on a representation contract and the rejections and deferral on physical and numerical reasoning that follows from the task's governing equations and discretization rather than from the surrogate's outputs, which is why the typed structure transfers and discriminates across tasks even on this converged model. This is a primary-scale second-task gate-discrimination result on the official MeshGraphNet architecture (240 checkpoint-by-trajectory cells), not an official-checkpoint or SOTA-accuracy airfoil result.

5.6. RQ1 operator-floor resolution sweep

The P1 discrete-divergence sweep on symmetric meshes, over nine resolutions, gives slope 0.984 with 95% CI $[0.975, 0.992]$ and $\mathrm{R}^2 = 0.9999$, matching O(h) (Figure 3). In addition to the slope, the *absolute* floor is closed-form for the concrete deployed-scale mesh: since the operator is exact on affine fields and the reference field is divergence-free, the measured floor equals the geometry-weighted second-order Taylor remainder exactly. A leading-order predictor from the analytic Hessian matches the measured floor to within 0.5% (1.337 vs 1.343), and a rigorous a-priori upper bound from the Hessian's global spectral norm dominates it at every resolution. The same floor reproduces on a genuinely different topology — an unstructured Delaunay triangulation of a jittered grid (worst triangle angle about 14 degrees) over the same domain and reference field — with log–log slope 0.983 (Student-t 95% CI $[0.953, 1.014]$, $\mathrm{R}^2 = 0.999$), a matched-resolution unstructured-over-structured floor ratio of 1.06 median and 1.33 maximum, so the numerical-decidability verdict does not flip between the two topologies. The gate thus rests on a derived floor, not an empirical estimate, for this mesh; the floor is empirically topology-stable across these two mesh families, while a general unstructured-mesh bound remains future work. This is why absolute conservation remains deferred while the reference-relative non-regression guard is reported. A classical flux-form finite-volume operator instead admits a fully executable conservation verdict (claim C34, supporting artifact).

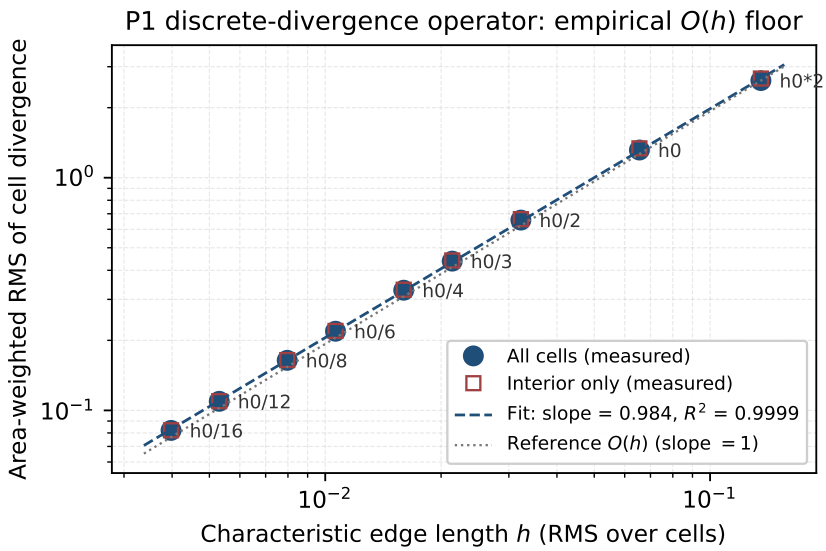

Figure 3: Operator-floor calibration for the P1 divergence measurement, whose log–log sweep gives the $O(h)$ floor used to decide whether absolute conservation is numerically admissible for the cylinder-flow mesh.

5.7. Coverage implication: fault-detection robustness across the multi-checkpoint roster

To check whether the seeded-fault detection result of Section 5.3 reproduces beyond a single checkpoint and to delimit where its detectors are structurally insensitive, the 10-mutant catalogue is replayed against the K=6 multi-checkpoint roster across five input-permutation seeds (30 trials per mutant per detector), and two predeclared severity dimensions are swept: NS_double_scale $s \in \{1.1, 1.25, 1.5, 2, 4\}$ and PC_zero_vy partial fraction $p \in \{0.25, 0.5, 0.75, 0.85, 0.9, 0.95, 0.99, 1.0\}$ (the canonical mutant is $p = 1.0$). Detection is decided by the same predeclared thresholds as Section 5.3 (node-permutation tolerance 10^{-5}, conservation ratio threshold 1.5, mirror-y relative-change threshold 0.5). Wilson 95% CIs are computed across the (SUT, seed) cells; raw outputs are committed (Table A1).

Aggregate reading and Phase-3 unified catalogue. The 5-of-10 union detection rate from C10 reproduces across S0–S3 (4/10 on S4/S5). By-class localization (continuity $\to$ boundary/scale; symmetry $\to$ physical-channel/adjacency) replicates across the cylinder checkpoint roster (its cross-SUT transfer is examined below), and two predeclared severity sweeps bound the insensitivity regions with Wilson 95% CIs: multiplicative output scaling stays undetected at every scale (0/6, CI $[0.00, 0.39]$), while partial v_y-zeroing is caught at every fraction up to $p = 0.99$ (6/6) and blind only at the exactly-uniform $p = 1.0$ (0/6), a measure-zero knife-edge symmetry blind spot that the adversarial mutants further delimit. This reflects the detectors' coverage geometry: each MR scores a single invariant, so a fault is caught only when it perturbs a measured invariant and is structurally invisible when it preserves all of them; closing the blind region calls for additional MRs probing the uncovered directions, not more mutants of the same kind. Phase 3 additionally emits a 60-entry unified fault catalogue (Table A1): 10 executed canonical MGN mutants, 2 executed adversarial MGN mutants, 24 closed-form output-level PINN probes, and 24 closed-form output-level FNO probes. The precision/recall summary (descriptive catalogue rates over non-independent checkpoint-by-seed cells, not independent-item estimates) is node-permutation 1.00/1.00, conservation 1.00/0.81, and mirror-y 0.94/0.55; the same artifact reports Cliff's δ and Wilcoxon signed-rank tests for the PINN MR-B ratios at $n = 15$ per PDE: Cliff's $\delta = 0.73$ (large) and Wilcoxon $p = 0.0084$ for diffusion-vs-Burgers MR-B violation/floor ratios. This inter-PDE test is a physics-magnitude comparison (Dirichlet-zero Burgers BCs predict near-zero boundary residuals while Neumann-zero-flux diffusion BCs permit larger boundary residuals, so the magnitude difference is expected by construction) and is *not* a gate-reliability test. The valid reliability statistic is the per-PDE Wilson CI on the MR-B pass rate: Burgers 13/15 ($[0.62, 0.96]$) and diffusion 7/15 ($[0.25, 0.70]$). The PINN/FNO probes are not retrained mutant checkpoints. Re-run with realistic mechanism-level surrogate bugs that are not tailored to the relations (boundary mis-scaling, renormalization, channel swap, region dropout, fixed Gaussian noise, spatial shift, sharpening, mode truncation; per-fault output perturbation recorded), the by-class structure emerges rather than being constructed: on both a spectral FNO and a pointwise Burgers PINN, detection partitions by whether a fault changes the conserved channel integral and whether it breaks the symmetry or translation relation. The single-relation cells are near-definitional, since the conservation relation is itself the test of whether the integral changed, so the load-bearing and falsifiable content is coverage completeness: a fault preserving every measured invariant is predicted in advance, from the fault operator alone, to be invisible to the whole relation suite, and on both architectures every such fault is detected by no relation, with no falsifying case. This blindness is independent of how badly the fault degrades the output — a transport fault corrupts the PINN field by 0.63 relative L2 yet stays invisible to every relation, while on the smoother FNO fields no invariant-preserving fault reaches a comparable magnitude, so the practical reach of the blind region is itself field-structure dependent. The trained-mutant MeshGraphNets and PointMLP evidence and the realistic-fault FNO/PINN evidence are not interchangeable; this is bounded cross-architecture evidence for a falsifiable coverage geometry, not generalized to all relations or architectures.

Adversarial mutants (R4). Two adversarial mutants test whether the blind spot is a subspace, not a point. A1 is magnitude-driven, with node-permutation 0/6 and conservation 0/6; A2 escapes every detector. These results bound the detector set rather than establishing validated localization.

Cross-SUT coverage and its falsifiable boundary. The seeded-fault stress tests suggest a bounded, falsifiable organizing principle, the validity-coverage duality: the same admissibility gate that decides whether an MR yields a meaningful verdict also fixes which faults that MR can detect, so coverage follows the admissible MR set. We test it on the primary-scale PhysicsNeMo airfoil (Section 5.5), with the same 10-mutant catalogue and predeclared thresholds, over five test trajectories (claim C36): the non-zero angle of attack makes mirror-y domain-inadmissible, so the gate excludes it and the mirror-only cylinder-flow fault coverage is absent on airfoil, while the one localization shared across both SUTs is continuity $\rightarrow$ normalization-scale (conservation still detects the gross normalization fault on all five trajectories; node-permutation detects none). The principle is refutable: coverage surviving exclusion of the only MR measuring the affected invariant, a fault caught while perturbing no admitted invariant, or a blind spot closed by same-kind mutants rather than new relation directions would each weaken it; here the cross-SUT mirror-y removal supports relation-specific exclusion, and the knife-edge sweep at $p = 1.0$ with the adversarial mutants delimit blind subspaces. The claim stays qualitative and fault-class level on two CFD SUTs, a falsifiable hypothesis rather than a calibrated coverage law or general theorem, and not a real-world defect-rate estimate.

Cross-program breadth. The Minimum-MR-SubSet sibling is used only as an external breadth audit: seven program types are inspected read-only and five CPU-only SUTs are replayed through this paper's gate and typed verdict (claims C39–C40). Appendix Appendix B gives the full results.

5.8. Cross-RQ boundary of the evidence

The canonical blocked list is narrowed but still active: the evidence does not support external-dataset or geometry-independent pass/fail rates, AeroGraphNet/DoMINO results, comparative superiority over baselines, general or real-world fault-detection rates, validated localization, runtime, reliability, or model accuracy claims.

5.9. Cross-RQ interpretation and discussion

5.9.1. Interpretation of the scoped evidence

The value of the current study is not that every MR finds a new fault beyond rollout accuracy. Retained MRs provide relation-level evidence under explicit transformations; when a violation or deferral occurs, the MR card and verdict rule help distinguish model inconsistency, relation-domain boundary, numerical tolerance issue, and inconclusive evidence.

A natural objection is that the admissibility predicate merely re-describes the three initial outcomes. The added runs answer this: the symmetric-mesh run uses the predicate out-of-sample and fails an admitted relation; rollout accuracy shows the mirror-y violation answers a different question from one-step error; and the primary empirical scope upgrade preserves these readings across a K=6 roster, three held-out trajectories, and larger denominators. The PINN roster, FNO primary workflow, and Minimum-MR-SubSet MGN/PINN reruns broaden applicability context while preserving the reliability and geometry-independent-rate boundary.

A second objection reads the deferred absolute-conservation relation as a gap in the end-to-end pipeline. The portfolio answers this directly: the conservation class produces full executable verdicts wherever its measurement floor is calibratable (FNO periodic conservation fails 24/24 against a calibrated floor; PINN reference-relative conservation passes 30/30).

The rubric averts two concrete misreadings: absolute discrete-divergence would have read as a conservation *pass* despite reference-field divergence, and asymmetric-mesh mirror-y would have read as a model fault rather than a geometric artifact.

5.9.2. Boundary of claims

The paper avoids four overclaims: that MRs always beat rollout accuracy; that NOETHER proves the physical validity of cylinder-flow MRs; that LLMs can automatically identify valid MRs; and that one MeshGraphNets family generalizes to all SciML surrogates. The safer claim is that a domain-validity-aware workflow makes MR identification and execution more auditable for a concrete class of SciML SUTs.

5.9.3. Implications for SciML testing

The scoped evidence shows how SciML testing can move from implicit expert checks to explicit MR assets that complement residuals, UQ, and accuracy by making transformations and verdict rules inspectable. The workflow targets a relation-indexed applicability map in relation space rather than residual space. We do not claim a completed applicability map in this paper; the present evidence is one bounded within-SUT point, and calibrated maps across SUTs, trajectories, and domain-violation scores remain future work.

When to use, and at what cost. The gate earns its overhead when a surrogate must be trusted out of distribution and a physically necessary relation can be both stated for the task and made numerically decidable for the deployed discretization; it complements rather than replaces accuracy and uncertainty monitoring, answering whether a declared relation holds rather than how large the error is (Section 2.6). Authoring an admissible MR costs domain-physics knowledge to state the relation and a one-time numerical-analysis step to establish the measurement floor, after which the MR card and its runner replay automatically, so the cost is a per-relation design investment rather than a per-input runtime price. A practitioner already running an in-band accuracy monitor gains most from a gate-admitted relation exactly where a relation violation would otherwise stay invisible, as the seeded-fault contrast shows for faults that leave one-step error within its baseline band while breaking a measured invariant (Appendix Appendix C).

6. Threats to Validity

We classify threats following Verdecchia et al. [46] and report against the Empirical Standards for software engineering research [47].

Construct validity. The main construct is not "model correctness" but validity-gated MR asset construction. A wrong physical premise, boundary-condition reading, output mapping, discrete operator, or tolerance could therefore change an admit/reject/defer decision. We mitigate this threat by requiring an MR card, explicit preconditions, typed verdicts, and an operator-floor argument before a relation is treated as executable evidence. The mitigation is still incomplete: the P1 divergence floor is derived for the deployed symmetric-mesh setting and is empirically stable on a second unstructured Delaunay topology but is not proven for arbitrary unstructured meshes, and the reference-relative conservation diagnostic remains a non-regression guard rather than an absolute conservation measurement. The seeded-fault catalogue also measures detector sensitivity to author-implemented mutants, so its rates do not estimate real-world defect prevalence or effectiveness.

Internal validity. SUT setup, checkpoint selection, random seeds, mesh preprocessing, normalization, and runtime nondeterminism may affect verdicts. The study records manifests, ledgers, and raw outputs, and it repeats the primary cylinder-flow checks across K=6 checkpoints, three held-out trajectories, same-task variants, PointMLP, and PhysicsNeMo. These repetitions reduce the risk that the pilot is a single-run artifact, but they do not make the cells independent population samples. For airfoil, the surrogate is trained to convergence (40 epochs, final loss near 0.14) but its one-step eval rollout remains high (relative L2 0.92); its evidence is read for typed gate discrimination, not model accuracy.

External validity. The evidence covers incompressible cylinder flow, compressible airfoil flow, bounded PINN/FNO Burgers and heat executions, and a read-only/executed cross-program breadth check from the Minimum-MR-SubSet sibling artifacts. This is enough to show that the gate is not a one-off cylinder-flow checklist, but it remains far from a general claim about SciML surrogates. Broader neural operators, mesh simulators, production CFD solvers, PINN architectures, datasets, geometries, and training regimes remain outside the evidence. The cross-program breadth check reuses the sibling's domain assets and mutant sets, so it supports only a qualitative coverage-geometry reading, not per-program reliability, new MR discovery, superiority, or real-world fault-detection rates.

Baseline and comparator validity. Rollout accuracy, residual/UQ-style diagnostics, generic MR templates, and LLM-assisted candidate generation are used as scope contrasts rather than defeated competitors. The LLM baselines depend on prompt design, model versions, and simulated raters; they are useful for showing why an admissibility gate is needed, but they are not a human-expert study and do not establish superiority over alternative MR-identification methods.

Conclusion validity. The study combines descriptive counts, Wilson intervals, bootstrap intervals, and exact nonparametric tests where the sampling unit supports them. Many reported cells still share checkpoints, trajectories, or generated cases, so they are not independent population trials. The primary MGN scope upgrade improves the weakest denominators within the stated scope, but the correct conclusion is bounded empirical utility for RQ4, not a calibrated reliability or coverage rate.

Reproducibility. Some workflows depend on old runtimes, public benchmark loaders, or non-redistributable checkpoints. The replication package therefore emphasizes scripts, manifests, run logs, claim ledgers, and available outputs. Reproducibility is strongest for artifact inspection and rerunnable CPU workflows, and weaker for exact recreation of every trained checkpoint environment. By re-run cost: the operator-floor sweep, MR-card validators, and typed-verdict logic replay on CPU in minutes from committed scripts alone; the cylinder-flow MGN pilot, K=6 roster, PointMLP, and FNO/PINN rosters replay on CPU from committed checkpoints and metric ledgers; PhysicsNeMo MGN and airfoil training require a GPU and the public DeepMind TFRecords; and the Minimum-MR-SubSet audit and rerun execute read-only from sibling fixtures at the cited commit.

7. Future Work

Future work should generalize the numerical admissibility gate beyond the deployed P1 symmetric-mesh setting to unstructured meshes, boundary treatments, and flux-form operators.

Second, the domain-violation axis should be calibrated across MR families. Here, domain distance is a per-relation interpretive coordinate, not a cross-MR score. A calibration protocol would normalize distances within relation families, define in-domain/near-boundary/out-of-relation-domain anchor cases, build relation-family calibration sets, and use independent domain experts or raters to validate categories, agreement, and residual comparability limits.

Third, the MR-card workflow should be evaluated on production-scale SUTs and datasets, including external aerodynamics, production graph simulators, longer rollouts, and multi-physics pipelines.

Fourth, the coverage implication should be tested with independent real-defect corpora and independently authored mutants, asking whether admitted MR sets predict externally collected defect classes.

Finally, MR-card authoring, precondition checking, and ledger generation need automation, while validity decisions remain tied to physical, numerical, and software evidence rather than model consensus.

8. Conclusion

This paper asked how physics-derived MRs for SciML surrogate software can be transformed into auditable, executable, and interpretable oracle-free V&V assets. The answer is a validity-gated workflow: a candidate relation is first checked for physical basis, domain preconditions, representation mapping, and numerical decidability; admitted relations are recorded as MR cards and executable assets; and outcomes are interpreted with typed verdicts rather than a single pass/fail label.

The empirical evidence is bounded but concrete. On cylinder-flow MeshGraphNets, the workflow separates a representation pass, an out-of-relation-domain mirror probe, a deferred absolute conservation relation, and a reference-relative diagnostic; the same readings are replayed across checkpoints, trajectories, and same-task architectures. On airfoil, PINN, and FNO subjects, the gate changes or preserves verdicts according to the relation's physical and numerical conditions. Seeded-fault stress tests further show that the admitted MR set helps explain detector blind spots in the studied settings, without implying general defect-detection rates.

The contribution is therefore methodological: physically meaningful SciML metamorphic tests require explicit validity conditions, executable assets, raw evidence records, and relation-level verdicts. This turns candidate MRs from informal expert intuitions into auditable V&V assets whose boundaries are visible to reviewers and practitioners.

Preprint

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CRediT authorship contribution statement

Meng Li: Conceptualization, Methodology, Software, Writing – original draft. **Xiaohua Yang:** Supervision, Formal analysis, Writing – review &

editing. **Jie Liu:** Investigation, Validation. **Shiyu Yan:** Data curation, Visualization.

Declaration of competing interest

The authors declare no conflict of interest.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used generative AI assistants for code scaffolding, prose copy-editing, and reference cross-checking. After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. No generative AI was used to fabricate, alter, or invent experimental data, results, or citations; every numerical claim is backed by a tracked artifact under `research_assets/runs/` and a claim-ledger entry.

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Data availability

The replication package contains the manuscript source, assets, ledger, and outputs, and is archived on Zenodo at `https://doi.org/10.5281/zenodo.20702952`. Minimum-MR-SubSet audit/rerun artifacts are under `research_assets/runs/` and cite commit `9ef862ec37335b4834d0a1fb38b4b613af702f34`. The MeshGraphNets checkpoints derive from the public DeepMind cylinder-flow benchmark; no proprietary data was used.

Appendix A. Claim-to-evidence map

Table A1: Compact claim-to-evidence map for the current manuscript.

Claim	Status	Evidence	Boundary
Domain-validity rubric	Method claim	Rubric asset and manuscript method section	Establishes an auditable screening rule; physical validity still depends on each relation's stated preconditions.
MR-card executable assets	Workflow claim	MR cards and validators	Some cards remain protocol assets pending matched SUT evidence.
Baseline admissibility contrast	Observed	Expert-MR, generic-MR, LLM-MR baselines and claim ledger	Quantifies an admissibility gap without ranking methods as competitors.
Three-arm complementarity / gate value	Observed	PointMLP three-arm report and detection-vs-accuracy metric ledger	MR suite 13/20 vs accuracy monitor 6/20 (2×2: 9/2/4/5, Wilson CIs); gate-admitted template 0% vs gate-rejected templates 100% baseline false positive; a measured gate value and complementarity, never a superiority ranking; consolidated across three converged SUTs (cylinder MGN 5/10 vs 3/10, airfoil 1 vs 2 disjoint, airfoil gate-rejected templates 4/4 firing on the correct SUT; C51).
Minimum-MR-SubSet external-scope audit	Observed secondary audit	External-scope audit JSON	external witness evidence from 70 real rows and three SciML/PDE true-fault-class PASS_WITNESS rows; does not add new primary SUT executions to this paper.
Minimum-MR-SubSet primary rerun	Observed external runtime rerun	ABD witness rerun report	PASS_WITNESS, kstar = 6, four active true fault classes, max signature rank 2, collapse false; real held-out cylinder-flow MGN runtime, not a second architecture or dataset.
Minimum-MR-SubSet PINN primary reruns	Observed second-SUT/PDE reruns	ABD PINN witness rerun reports	trained Burgers2D and Diffusion2D PINN witnesses, PASS_WITNESS, kstar = 1 each, five active true fault classes each; one-seed witnesses, no cross-SUT rate.
Node-permutation sanity check	Observed pilot	Node-permutation metric ledger	One representation-contract path on one pilot case.
Conservation diagnostic	Diagnostic; absolute relation deferred	Conservation metric ledger and report	Reference-relative guard only; absolute conservation remains deferred.
Mirror-y OOD stress (PC6-mirror-y-ood-stress)	Observed bounded pilot	Mirror-y rate metric ledger and claim ledger	Failed on 10 of 10 recorded eval frames; not a reliability, accuracy, baseline, multi-SUT, exact-symmetry, or geometry-independent claim.

Claim	Status	Evidence	Boundary
Primary empirical scope upgrade	Observed	Primary-scope report and per-trajectory/checkpoint manifests	K=6 x 3 trajectories x 10 mirror-y OOD-stress grid fails 180/180; K=6 x 3 trajectories x 9 conservation-transition grid passes 162/162; K=6 x 3 exact-symmetric-mesh input grid fails 18/18; not a single-source-trajectory estimate and not a cross-SUT rate.
LLM role boundary	Process boundary	Method and ethics sections	LLMs organize candidates; the rubric decides validity.
Rollout-accuracy diagnostic	Observed	Rollout-accuracy metric ledger	Same-SUT one-step median relative L2 0.0216; mirror-y is ∼34x larger.
Exact mirror-y symmetric mesh	Observed	Symmetric-mesh metric ledger	Exact relation admissible and fails (rel L2 1.10) on a synthetic OOD mesh; binary equivariance evidence.
PC10-seeded-fault-detection	Observed	Seeded-fault metric ledger	Detector stress test: MRs catch 5/10 injected mutants with class-specific response patterns.
Multicheckpoint replication	Observed	E1 aggregate and per-SUT manifests in `multicheckpoint/`	K=6 checkpoints of one MeshGraphNets family; within-family replication.
Operator-floor resolution	Observed	Operator-floor sweep report	Log–log slope 0.984, 95% CI $[0.975, 0.992]$ (interior 0.989), $R^2 = 0.9999$ over nine resolutions; reproduced on a second unstructured Delaunay topology (slope 0.983, floor ratio 1.06 median / 1.33 max), calibrating the admissibility predicate across two mesh topologies.
Fault-detection robustness	Observed	E3 robustness report; Phase-3 unified catalogue	30-trial Wilson CIs per MGN mutant plus a 60-entry unified fault catalogue (10 canonical MGN + 2 adversarial MGN + 24 closed-form PINN output-level probes + 24 closed-form FNO output-level probes), by-detector precision/recall with Wilson CIs, and Wilcoxon/Cliff effect-size tests; with realistic non-tailored faults the by-class structure emerges as a two-by-two partition (changes integral × breaks symmetry) on FNO and PINN, and a falsifiable coverage-completeness holds — every invariant-preserving fault is invisible to the whole suite with no falsifying case, blind even at 0.63 relative L2 on PINN; PINN/FNO probes are closed-form output-level probes.

Claim	Status	Evidence	Boundary
S4/S5 variant primary workflow	Observed	Variant report and raw ledgers	Same-domain wider/deeper MGN variants: 2/2 node-permutation passes, 60/60 mirror OOD failures, 54/54 conservation-diagnostic passes, 6/6 exact-symmetric failures; not PhysicsNeMo/EchoWave or cross-dataset reliability evidence.
PointMLP cylinder primary workflow	Observed	PointMLP report, checkpoint, and raw ledgers	Newly trained non-MGN row-wise coordinate network: 9/9 node-permutation passes, 10/10 mirror OOD failures, 9/9 conservation-diagnostic passes, 3/3 exact-symmetric failures, median rollout rel L2 0.0298; not PhysicsNeMo/EchoWave or production CFD evidence.
PhysicsNeMo MGN Object-A smoke workflow	Observed smoke-subset production-framework execution	PhysicsNeMo smoke report, newly trained checkpoint, raw outputs, and metric ledgers	NVIDIA PhysicsNeMo `MeshGraphNet` executes on a first-record official DeepMind `cylinder_flow` smoke subset: node permutation passes (relative L2 0.0), mirror-y is OOD stress, and conservation is reference-relative diagnostic only; not full production-scale PhysicsNeMo, AeroGraphNet, or DoMINO evidence.
FNO primary workflow upgrade	Observed	FNO workflow report, per-SUT ledgers, and raw source/follow-up outputs	Six trained FNO-2D checkpoints over Burgers/heat: 24/24 translation passes, 24/24 conservation failures under a periodic discrete-conservation MR, and 6/6 Dirichlet-boundary translation rejections; not only admissibility evidence.

Appendix B. Cross-program breadth

Cross-program generalization. Reusing the committed Minimum-MR-SubSet kill matrices, the coverage-geometry reading reproduces across seven program types in three families: neural surrogates, classical numerical solvers (wave, point-kinetics, Burgers), and a production Monte-Carlo physics code (OpenMC). For each, fault coverage is the union of its admissible MR set's per-class coverage; structural blind spots are present in six of the seven; and the class-to-class mapping is program-specific, extending the SUT-specificity finding from two CFD SUTs to seven program types. Detection rate tracks the admissible MR set rather than being uniform (minimal MR sets reach 10–20% with large structural blind spots, richer sets up to 100%), which is the coverage-generation structure the principle predicts. This is a read-only generalization check from committed evidence, not an end-to-end execution of this pipeline on each program (claim C39).

End-to-end cross-program execution. Beyond that read-only reuse, this paper's pipeline is executed end-to-end on five CPU-only SUTs imported read-only from the same sibling: four classical solvers (heat parabolic, wave hyperbolic, point-kinetics stiff-ODE, Burgers conservation law) and the sibling's OpenMC subject run with the real OpenMC Monte-Carlo k-eigenvalue solver (single-energy-group infinite medium, verified against the closed-form k_∞), applying the four-condition gate, the typed verdict, and the coverage map to each program's own domain MRs and seeded mutants. The gate is no rubber stamp: across the five it admits 38 candidate relations and rejects 7 (two on wave, three on point-kinetics, two on OpenMC) at the output-mapping condition, where a one-sided monotonicity relation is not falsified by a scrambled-observable probe, and defers none. Each admitted relation holds on the correct program (the in-domain, consistent region of the typed verdict) while the mutant runs populate the SUT-inconsistency region; every program keeps structural blind spots once the sibling's declared-equivalent mutants are set aside, and the operator-class-to-coverage mapping is program-specific. Detection tracks the admissible set rather than being uniform (full-set mutation score 0.75 on heat, 0.20 on the other three solvers, and 0.56 on OpenMC, for the sibling's committed mutant sets). The MRs and mutants are the sibling's; only the gate and typed verdict are newly executed here, so this asserts no per-program reliability, no superiority, no new MR contribution, and, for OpenMC, no continuous-energy or whole-core reactor-physics result (claim C40).

Appendix C. Secondary baselines, external-scope audit, and cross-family executions

Appendix C.1. RQ4 secondary baselines and external-scope audit

This subsection reports the Secondary baseline and external-scope audit; LLM baselines are secondary exploratory scope contrasts, not competitors (expert-LLMs proposed 24 unique candidates of which rubric voting retained 4 with no novel family; generic templates admitted 3/13; one-shot rater agreement Fleiss kappa 0.077, a low agreement that concerns only the LLM candidate-screening step and not the deterministic operator-floor admissibility condition, which requires no rater agreement). The load-bearing secondary result is a measured gate value on the converged PointMLP SUT:

over a 20-fault catalogue the validity-gated MR suite and a rollout-accuracy monitor are complementary (suite 13/20, monitor 6/20; a 2×2 split of MR-only 9, accuracy-only 2, both 4, neither 5), and the single gate-admitted generic template raises no baseline false positive on the fault-free SUT (0%) whereas all six gate-rejected templates fire on it (100%), so the admissibility gate is what removes the false-alarming detectors; this is complementarity and a measured gate value, with descriptive Wilson CIs and no hypothesis test. The same contrast holds across the converged SUTs (cross-SUT consolidation, Table A1): on every SUT the validity-gated MR arm catches relation violations the accuracy monitor leaves in band (cylinder-flow MeshGraphNets 5/10 vs 3/10; the low-fidelity airfoil 1 vs 2, where the two arms are disjoint), and on the airfoil too the gate-rejected templates fire on the correct SUT while the admitted detector stays clean (4/4, including the inadmissible mirror-y), the airfoil's reduced coverage following exactly from its non-zero angle of attack making mirror-y inadmissible. The Minimum-MR-SubSet audit/rerun is likewise an external witness (70 real ABD instances, three SciML/PDE PASS witness rows, one held-out cylinder-flow MGN rerun) that adds artifact-backed scope context without adding a second primary architecture/dataset.

Appendix C.2. RQ4 Cross-family PINN extension and FNO upgrade

The cross-family executions test two claims the cylinder-flow study cannot test by itself: that the admissibility predicate is family-agnostic (the same four conditions, applied unchanged, produce typed decisions on pointwise PINNs and spectral FNOs), and that the conservation MR family (b) yields full executable verdicts wherever its floor is calibratable: the FNO periodic conservation MR fails 24/24 against a calibrated case-level floor and the PINN reference-relative conservation passes 30/30, while the MGN open-boundary absolute variant is the one member of the family the gate correctly refuses. The deferral on cylinder flow is the fail-closed gate discriminating rather than a missing verdict.

To check whether the workflow is MeshGraphNets-specific, we use a K=15 PINN roster: fifteen Burgers and fifteen heat seeds. **MR-A** remains *vacuous by construction* for a pointwise MLP. The two non-trivial MRs do the cross-family work: **MR-B** passes on 13/15 Burgers seeds (violation/floor ratio mean 0.712, bootstrap CI [0.583, 0.875]; Wilson pass-rate CI $[0.62, 0.96]$) and is mixed on heat with 7/15 passing (mean 1.495, CI [1.142, 1.897]; Wilson pass-rate CI $[0.25, 0.70]$), two seeds near the residual-floor boundary fail on Burgers, while the Neumann-zero-flux heat BCs predict larger boundary residuals and yield a lower pass rate by construction; **MR-C** passes on all 15/15 of both PDEs (Wilson CI $[0.80, 1.00]$), with Burgers mean ratio 1.007 (bootstrap CI [0.997, 1.017]) and heat mean ratio 1.006 (CI [0.988, 1.022]). This is a two-PDE PINN seed roster rather than a PINN-vs-MGN benchmark or a generality claim across PINN architectures, PDEs, or training regimes.

The **FNO primary workflow upgrade** converts the earlier FNO roster into a second trained primary execution. Six small FNO-2D checkpoints cover Burgers and heat. For each checkpoint and four held-out generated periodic finite-difference cases, the workflow records rubric decisions, source and follow-up tensors, mapped outputs, metric ledgers, and relation verdicts. Periodic integer translation is admitted and yields **24/24 translation passes** (maximum relative-L2 violation below 10^{-5}). The **periodic discrete-conservation MR** is admitted-with-reference-floor because the finite-difference target supplies a case-level channel-sum drift floor; the trained FNO outputs exceed that calibrated floor on **24/24 conservation failures**. The Dirichlet translation candidate is rejected for 6/6 SUTs because it changes the boundary-value problem and is not executed as an exact MR. This is **not only admissibility evidence**: it is a full rubric-to-verdict FNO execution with raw source/follow-up outputs and per-case ledgers, while remaining outside cylinder-flow evidence, performance benchmarking, reliability, and broad neural-operator generalization. Separately, the FNO admissibility roster is extended to $K = 15$ seeds per PDE so the periodic-translation admission decision is adequately powered: it is admitted on 15/15 per PDE (Wilson 95% CI $[0.80, 1.00]$), with near-machine-precision violation magnitudes (Burgers mean 4.5×10^{-8}, heat 1.4×10^{-7}).

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